```
100 'CHEBYSHEV FILTER-  COEFFICIENT CALCULATION
110 '
120                                 'INITIALIZE VARIABLES
130 DIM A[22]                       'holds the "a" coefficients
140 DIM B[22]                       'holds the "b" coefficients
150 DIM TA[22]                      'internal use for combining stages
160 DIM TB[22]                      'internal use for combining stages
170 '
180 FOR I% = 0 TO 22
190   A[I%] = 0
200   B[I%] = 0
210 NEXT I%
220 '
230 A[2] = 1
240 B[2] = 1
250 PI = 3.14159265
260                                 'ENTER THE FILTER PARAMETERS
270 INPUT "Enter cutoff frequency  (0 to .5):       ", FC
280 INPUT "Enter  0 for LP,  1 for HP filter:       ", LH
290 INPUT "Enter percent ripple    (0 to 29):       ", PR
300 INPUT "Enter number of poles (2,4,...20):       ", NP
310 '
320 FOR P% = 1 TO NP/2              'LOOP FOR EACH POLE-ZERO PAIR
330   '
340   GOSUB 1000                    'The subroutine in Fig. 33-8
350   '
360   FOR I% = 0 TO 22              'Add coefficients to the cascade
370     TA[I%] = A[I%]
380     TB[I%] = B[I%]
390   NEXT I%
400   '
410   FOR I% = 2 TO 22
420     A[I%] = A0*TA[I%] + A1*TA[I%-1] + A2*TA[I%-2]
430     B[I%] =      TB[I%] - B1*TB[I%-1] - B2*TB[I%-2]
440   NEXT I%
450   '
460 NEXT P%
470 '
480 B[2] = 0                        'Finish combining coefficients
490 FOR I% = 0 TO 20
500   A[I%] = A[I%+2]
510   B[I%] = -B[I%+2]
520 NEXT I%
530 '
540 SA = 0                          'NORMALIZE THE GAIN
550 SB = 0
560 FOR I% = 0 TO 20
570   IF LH = 0 THEN SA             = SA  +  A[I%]
580   IF LH = 0 THEN SB             = SB  +  B[I%]
590   IF LH = 1 THEN SA             = SA  +  A[I%] * (-1)^I%
600   IF LH = 1 THEN SB             = SB  +  B[I%] * (-1)^I%
610 NEXT I%
620 '
630 GAIN = SA / (1 - SB)
640 '
650 FOR I% = 0 TO 20
660    A[I%] = A[I%] / GAIN
670 NEXT I%
680 '                               'The final recursion coefficients are
690 END                             'in A[ ] and B[ ]
```

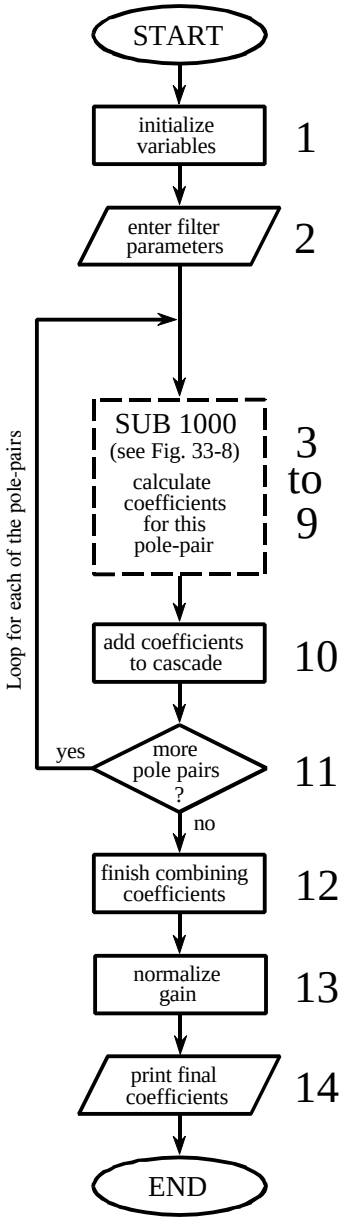

FIGURE 33-7
Chebyshev-Butterworth filter design. This program was previously presented as Table 20-4 and Table 20-5 in Chapter 20. Figure 33-8 shows the program and flowchart for subroutine 1000, called from line 340 of this main program.